Project Proposal Template

Project: CardioScope: Cardiovascular Disease Risk Detection & Clinical Analytics Dashboard

Objective:. To examine patient health data, spot the main early warning signs of heart disease, and track key health indicators so doctors can easily identify and protect high-risk patients early.

Description: Heart disease is one of the leading causes of death worldwide, but catching it early saves lives. This project takes clinical health records—such as blood pressure, cholesterol, blood sugar, resting heart rate, age, and lifestyle habits—and organizes them into a clear system. The team will clean the raw data, uncover the most critical danger signs, calculate clear health indicators (KPIs), and build an interactive, easy-to-read dashboard that helps medical teams spot at-risk individuals before emergencies happen.

Tools & Design: excel, power BI, Tableau , python, SQL

Technologies to Use:

Data Processing & Analysis: Python ,SQL.

Dashboard & Visualization: Power BI , Tableau, Excel.

Collaboration: GitHub / Google Drive for code and document sharing.

Team Members & Responsibilities:

1. **Data Sourcing & Cleaning Specialist:** Finds reliable patient datasets, cleans missing or corrupted entries, removes impossible values, and prepares the clean master file.
2. **Statistical & Pattern Analyst:** Analyzes patient trends, identifies which risk factors (like smoking, age, or high blood pressure) correlate strongest with heart disease.
4. **KPI & Health Metrics Specialist:** Researches medical benchmarks to design clear health scorecards and warning thresholds (e.g., blood pressure risk levels, high-risk patient percentages)
5. **Dashboard & Visual Designer:** Designs and builds the interactive dashboard screens, filters, and charts so non-technical users and doctors can explore the data easily
6. **Project Coordinator & Report Lead:** Coordinates weekly tasks across the team, validates that findings match medical facts, and leads the writing of the final report and presentation slides.

Week 1:Data Gathering, Setup & Initial Cleaning

- Member 1: Collect raw heart disease datasets (e.g., UCI Heart Disease, Kaggle health records) and set up the shared project folder.
- Member 2: Run an initial check on the data to find missing information, odd numbers (like negative blood pressure), and duplicate rows.
- Member 3: Research standard medical ranges for cholesterol, blood sugar, and blood pressure to create a reference guide for the team.
- Member 4: Sketch simple draft layouts (wireframes) showing how the final screens and charts should look.
- Member 5: Organize team communication, outline the project charter, and document data definitions and project goals.

Deliverables:

- Shared folder with raw datasets safely stored.
- First cleaned version of the dataset with errors removed.
- Medical reference guide outlining normal vs. dangerous health ranges.
- Initial paper or digital sketches of the dashboard layout.

Week 2: Deep Data Cleaning & Pattern Discovery

- **Member 1:** Finalize the cleaned dataset, ensure all column types match, and save the final table ready for analysis.
- **Member 2:** Create charts showing patterns—such as how heart disease rates change across age groups, genders, and cholesterol levels.
- **Member 3:** Select the exact list of 5–7 main KPIs to track (e.g., High-Risk Rate, Average Blood Pressure by Age Group, High Cholesterol Rate).
- **Member 4:** Set up the dashboard file, import the clean data, and establish basic card views and color schemes.
- **Member 5:** Review findings against the medical reference guide to verify patterns make clinical sense; draft the Week 2 progress report.

Deliverables:

- Fully cleaned master dataset ready for reporting.
- Visual summary report showing key graphs and early warning patterns.
- Approved list of KPIs with simple definitions and calculation methods.
- First draft of the project report methodology section.

Week 3: KPI Calculation & Dashboard Building

- **Member 1:** Assist with calculating complex metrics and ensure data updates smoothly into the dashboard tool.
- **Member 2:** Group patients into risk categories (e.g., Low, Moderate, High Risk) based on their combined health markers.
- **Member 3:** Write the formulas inside the dashboard tool (using Excel formulas or Power BI measures) to compute all KPIs live.
- **Member 4:** Build the interactive charts, adding easy buttons and drop-down filters (e.g., filter by age bracket, gender, smoker status).
- **Member 5:** Test the dashboard buttons and numbers for accuracy; begin assembling the final written report.

Deliverables:

- Working interactive dashboard prototype showing live KPI scorecards and trend charts.
- Patient risk category breakdown (count and percentage of patients in each risk tier).
- Consolidated draft of the main project report.

Week 4: Final Testing, Polishing & Presentation

- **Member 1:** Organize all project files, scripts, and cleaned datasets into a neat, final submission folder.
- **Member 2:** Double-check all final numbers, percentages, and charts to guarantee complete accuracy.
- **Member 3:** Write a short "How to Read the KPIs" user guide to explain what each metric means for everyday healthcare decisions.
- **Member 4:** Polish the visual design of the dashboard (clean labels, clear contrast, easy-to-read text).
- **Member 5:** Finalize the complete project report and create the slide deck for the graduation presentation.

Deliverables:

- Completed, fully clickable Heart Disease Analytics Dashboard.
- User-friendly metric guide explaining the KPIs.
- Final Graduation Project Report.
- Professional presentation slide deck ready for delivery.